I PERIODIC TEST : 2025-26

CLASS - X (CBSE)

MATHEMATICS

Maximum Marks : 50
Time allowed : 2 hours

General Instructions:

(i) All question are compulsory.
(ii) This question paper comprises of four sections A, B, C and D.
(iii) Section – A, contains questions number 1 to 5 of one mark each.
(iv) Section – B, contains questions number 6 to 10 of two marks each.
(v) Section – C, contains questions number 11 to 15 of three marks each.
(vi) Section – D, contains questions number 16 to 20 of four marks each.
(vii) Internal choice is also given in some questions.

SECTION - 'A' [1×5=5]

Q.1. If the HCF of 52 and 91 is expressible in the form 10m+3 then the value of m is :
(a) 1 (b) 2 (c) 3 (d) 4

Q.2. If the sum of zeroes of the polynomial $p(x) = 2x^2 - k\sqrt{2}x + 1$ is $\sqrt{2}$, then value of k is :
(a) $\sqrt{2}$ (b) 2 (c) $2\sqrt{2}$ (d) $\frac{1}{2}$

Q.3. The pair of linear equation y = 0 and y = -6 has :
(a) A unique solution
(b) No solution
(c) Infinitely many solution
(d) Only solution (0, 0)

Q.4. The distance between points (-5, 4) and (1, 0) is :
(a) $2\sqrt{13}$ units (b) $\sqrt{13}$ units
(c) $4\sqrt{2}$ units (d) $2\sqrt{2}$ units

Q.5. Probability of having 53 Mondays in leap year is :
(a) $\frac{2}{365}$ (b) $\frac{2}{7}$
(c) $\frac{1}{365}$ (d) None of these

SECTION - 'B' [2×5=10]

Q.6. If α and β are zeroes of quadratic polynomial $5x^2 + 5x + 1$ then find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$.

Q.7. Find the ratio in which the point P(-4, 6) divides the line segment joining the points A(-6, 10) and B(3, -8).

Q.8. For what value of k, the pair of lines equations 2x+5y = k and kx + 15y = 18 has infinitely many solutions?

Q.9. The HCF of two numbers is 119 and their LCM is 11781. If one of the numbers is 1071, find the other.

OR

3 bells ring at an internal of 4, 7 and 14 minutes. All three bell ring at 6 a.m. When the three bells will ring together next?

Q.10. Find the mode of following data :

Class Internal	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70
Frequency	5	26	80	104	98

SECTION – 'C' [3×5=15]

Q.11. Two dice are thrown simulataneously. Find the probability of :
(i) getting a sum greater then 9
(ii) getting a doublet of even numbers
(iii) getting product of numbers on the dice is 25

Q.12. Show that $\frac{2+3\sqrt{2}}{7}$ is an irrational number, given that $\sqrt{2}$ is an irrational number.

Q.13. Find the zeroes of a quadratic polynomial $x^2 + 10x + 21$ and prove the relation between the zeroes and coefficients.

Q.14. Find the values of x for which the distance between the points A(x, 2) and B(9, 8) is 10 units.

Q.15. Find the mean of given data by using assumed mean method.

Class Internal	100 – 120	120 – 140	140 – 160	160 – 180	180 – 200
Frequency	8	3	8	6	5

OR

If the median of given data is 33, find the value of m.

Class Internal	0 – 15	15 – 30	30 – 45	45 – 60	60 – 75	75 – 90
Frequency	17	35	40	18	m	2

SECTION – 'C' [4×5=20]

Q.16. Prove that $\sqrt{3}$ is an irrational number.

Q.17. Show that the points (7, 10), (–2, 5) and (3, –4) are vertices of an isosceles right triangle.

Q.18. Solve thje following system of linear equations graphically -
$2x + y = 6$, $x - 2y + 2 = 0$

OR

A two digit number is such that the sum of its digits is 11. When the digits are reversed, the resulting number increases by 27. Find the original number.

Q.19. Cards marked with the numbers 1 to 100 are placed in a box. One card is drawn from this box. Find the probability that the number on the card is :
(i) a perfect square number
(ii) a prime number less than 20
(iii) an even number
(iv) a multiple of 20

CASE STUDY QUES.

Q.20. Pankaj's father gave him some money to buy mangoes from the market at the rate of, $p(x) = x^2 - 24x + 128$, α and β are zeroes of p(x).
Based on the above information, answer the following questions.
(i) What will be the shape of graph of p(x) ?
(ii) Find the value of α and β .
(iii) Find the value of $(\alpha+\beta)+\alpha\beta$

OR

What is the value of $\frac{1}{\alpha^2}+\frac{1}{\beta^2}$

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I PERIODIC TEST 2025- 2026
CLASS - X (CBSE)
SUBJECT - SOCIAL SCIENCE

Time : 2 hrs **M.M: 50**

General Instructions:

(i) The question paper is divided into six sections, i.e. section A, B, C, D, E and F.
(ii) All questions are compulsory.
(iii) Questions from section A are objective type questions. Each question carries 1 mark.
(iv) Questions from section B are very short answer type questions. Each question carries 2 marks.
(v) Questions from section C are short answer type questions. Each question carries 3 marks.
(vi) Questions from section D are long answer type questions. Each question carries 5 marks.
(vii) Question from section E are case- based question, with three sub- question and are of 4 marks each.
(viii) Question from section F is map based question carrying 2 marks. Attach the map of India along with the answer script.

SECTION- A

(Multiple Choice Questions)

Q.1 Which of the following is a renewable resource? [1]
a. Coal b. Water
c. Petroleum d. Natural Gas.

Q.2 Belgium took some measures to solve its problem. Read the following measures and choose the correct option: [1]
I. Formation of community government.
II. Following majoritarian policy.
III. By providing equal representation of language groups.
IV. By providing additional powers to local governments.
Options:
a. Only I, II and III are correct. b. Only I, III and IV are correct.
c. Only I, II and IV are correct. d. Only II, III and IV are correct.

Q.3 Read the following statements regarding the creation of states on the basis of language in India and choose the correct options: [1]
I. Formation of states on the basis of language made the country more united and stronger.
II. Formation of states on the basis of language made administration easier.
III. Formation of states on the basis of language has weakened the federal structure.
IV. Formation of states on the basis of language led to the development of Indian languages.
Options:
a. Only I,II and III are correct. b. Only II, III and IV are correct.
c. Only I, II and IV are correct. d. Only I, III and IV are correct.

Q.4 Subject like computer software comes in the [1]
a. Union List b. State List
b. Concurrent List d. Residuary List

Q.5 The idea of La Patrie means: [1]
a. Fatherland b. Motherland
c. Citizen d. Monarchy

Q.6 There are two statements. Assertion (A) and Reason (R). Answer these questions selecting the appropriate option given below. [1]
a. Both A and R are true and R is the correct explanation of A.
b. Both A and R are true but R is not the correct explanation of A.
c. A is true but R is false.
d. A is false but R is true.
Assertion (A) The success of federalism in India depends on more than just constitutional provisions.
Reason (R) Political parties must respect the autonomy of state governments.

Q.7 Black soil are common in- [1]

a. Deccan trap region b. Kashmir Valley

c. Ganga Valley d. Northern Plains

Q.8 What mark Greece as an independent nation [1]

a. Treaty of constantinople, 1832 b. Treaty of Vienna, 1815

c. Treaty of Versailles, 1871 d. Act of union, 1707

Q.9 Which state of India has the maximum area under permanent forest? [1]

a. Haryana b. Himachal Pradesh

c. Punjab d. Madhya Pradesh

SECTION -B

(Very Short Answer Type Quesstions)

Q.10 Mention any two benefits of dams.- [2]

Q.11 Why is power sharing good for democracy? Mention any two reasons. [2]

Q.12 How is over irrigation responsible for land degradation in Punjab? [2]

Q.13 Differentiate between unitary and federal form of Government with examples. [2]

SECTION - C

(Short Answer Type Questions)

Q.14 "The law enacted in 1956 and the other constitutional provisions led to the major conflicts between the two communities in Sri Lanka." Support the statement. [3]

Q.15 Explain any three reasons for the rising importance of tertiary sector in India. [3]

Q.16 Explain the Napoleonic Code of 1804. [3]

Q.17 Explain any three indicators that are used to measure the development of a country. [3]

SECTION -D

(Long Answer Type Questions)

Q.18 Explain the main features of alluvial soil. [5]

OR

What is soil erosion? What are the main causes of soil erosion?

Q.19 Why it is said that the 1830s were the years of great economic hardship in Europe? Explain [5]

OR

Describe the main clauses of the Treaty of Vienna of 1815.

SECTION - E

(Case Based Questions)

Q.20 **Read the following passages and answer the questions that follow.**

When power is taken away from Central and State government and given to local government, it is called decentralization. The basic idea behind decentralization is that there are a large number of problems and issues which are best settled at the local level. People have better knowledge of problems in their localities. They also have better ideas on where to spend money and how to manage things more efficiently. Besides, at the local level it is possible for the people to directly participate in decision making. This helps to inculcate a habit of democratic participation. Local government is the best way to realise one important principle of democracy, namely local self- government. The need for decentralisation was recognized in our Constitution. Since then, there have been several attempts to decentralize power to the level of villages and towns. Panchayats in villages and municipalities in urban areas were set up in all the States. But these were directly under the control of state governments. Elections to these local governments were not held regularly.

20.1 Explain Decentralization in democracy. [1]

20.2 Explain the importance of local self government in democracy. [1]

20.3 Describe any two steps taken by Indian government for decentralization. [2]

Q.21 For comparing countries, their income is considered to be one of the most important attributes. Countries with higher income are more developed than others with less income. This is based on the understanding that more income means more of all things that human beings need.

Whatever people like, and should have, they will be able to get with greater income. So, greater income itself is considered to be one important goal. Now, what is the income of a country? Intuitively, the income of the country is the income of all the residents of the country. This gives us the total income of the country.

However, for comparison between countries, total income is not such an useful measure. Since, countries have different populations, comparing total income will not tell us what an average person is likely to earn. Are people in one country better off than others in a different country? Hence, we compare the average income which is the total income of the country divided by its total population. The average income is also called per capita income. In World Development Reports, brought out by the World Bank, this criterion is used in classifying countries.

Countries with per capita income of US$ 49,300 per annum and above in 2019, are called rich countries and those with per capita income of US$ 2500 or less are called low- income countries. India comes in the category of low middle income countries because its per capita income in 2019 was just US$ 6700 per annum. The rich countries, excluding countries of Middle East and certain other small countries, are generally called developed countries.

21.1 Do you think that "Countries with higher income are more developed than others with less income." [1]

21.2 What is the criterion used by the world bank in classifying the countries? [1]

21.3 Why are the middle- east countries excluded from the list of developed countries? [2]

Q.22 On the outline map of India, locate and label the following. [3]

i. Identify Major Soil Type
ii. The tallest dam in India-
iii. Hirakud

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Tear here

I PERIODIC TEST- 2025-26
CLASS-X (CBSE)

Time : 2 hrs. **Subject- English Language & Literature** **M.M.: 50**

General Instructions:

i. *This question paper contains three Sections- A (Reading Skills, Section- B (Grammar & Writing Skills and Section- C Literature.*

ii. *Attempt questions based on specific instructions for each part.*

SECTION- 'A' (READING SKILLS)

Q.1 Read the following text :

1. In the classroom, broadening your thinking can mean emphasizing something called problem based learning. In this approach, a teacher presents a problem or question with no clear or obvious solution. Students are then asked to think broadly about how to solve it.
2. Problem based learning can help students think like scientists, Wallace says. He cites an example from his own classroom. Last fall, he had students read about fruit flies that lack an enzyme- a molecule that speeds up chemical reactions to break down alcohol. "I had seven groups of students, and got seven different ways to measure", he says. "That's what I would call creativity in a science class."
3. "Creativity means taking risks and not being afraid to make mistakes", adds Andrews. In fact, she and many educators agree, when something comes out differently than expected, it provides a learning experience. A good scientist would ask "Why?" She says, and "What's happening here?"
4. Talking with others and teamwork also help with associative thinking- allowing thoughts to wander and freely associating one thing with another- that DeHaan says, contributes to creativity. Working on a team, he says, introduces a concept called distributed reasoning, sometimes called brainstorming. This type of reasoning is spread out and conducted by a group of people.

Answer the following questions, based on the passage above.

a. Why does the writer say that problem- based learning can 'broaden one's thinking'? [1]

i. It can lead to the expansion of the human brain literally.
ii. It can lead to creative resolution of an issue at hand.
iii. It can lead to people working in a pedantic manner.
iv. It can lead to people working independently.

b. Complete the sentence appropriately: [1]
DeHaan lays deep emphasis on the benefits of associative thinking in science and emphasises that its maximum benefits can be drawn out _____

c. Based on your reading of the passage, define creativity as mentioned in the passage in about 30-40 words. [2]

d. Which of these explanations does not relate to the concept of distributive reasoning? [1]

i. It is synonymous with brainstorming.
ii. It is a part of associative thinking.
iii. It is prevalent among the scientific community only.
iv. It requires a team to conduct it successfully.

e. Which word in paragraph 1 is synonymous to 'easily clear'? [1]

Q.2 Read the passage given below:

1. Reading Comprehension (RC, as it is normally called) is the most peculiar section in almost all scholastic, entrance and employment tests. The skills in RC make a lot of difference to one's chances of good grades/ selection. Most students find it difficult to tackle topics that are diverse from the field they are in or they are comfortable with. So one needs to develop a taste for even the most obscure and boring topic on this planet. For success in RC one should be able to understand. And even with an average speed one can succeed if one implements the strategies.

2. Broadly speaking, RC passages can be classified in a few categories. Fact based RC is the simplest form of RC. These types of passages have lot of information in the form of names, numbers etc. In this type of passage one should read very fast.
3. Don't try to memorize any facts, numbers or names etc. In fact there is no need to even remember them. Just make yourself familiar with the structure of the passage. Just see in which paragraph author is talking about what. Mark it. Then when you go to the questions, identify in which paragraph information regarding that question is mentioned. Go to that paragraph, read the numbers, names etc. and mark the answer.
4. Inference based RC is the toughest form of RC. Here the passage is fairly tough to understand. This includes passages on topics like Religion, Spirituality, Philosophy, etc. Most of the students will not be comfortable attempting these passages at least in RC. The reading speed is fairly slow in this type of passages. The way to master this type of passages is to read them again and again while practising.
5. Topic based RC includes passage on any particular topic like economics, astrology, medical science, etc. Generally, what makes these passages tough is usage of technical terms. If a topic is new to us then presence of technical term scares us even if they are defined in the passage. For success in this type of passages, we need to have a fair understanding of the definition of the term if it is defined in the passage.
6. Reading passage first and then questions is the most popular strategy for RC. While answering the question you may come back to the passage to find answer as you have just read the passage initially and not crammed it. But you should not come back for each and every question. If you come back for majority of questions then you haven't read passage properly. The key to success for this strategy is that you should understand the passage very well.

On the basis of your understanding of the passage answer the following questions:- [6]

i. Why do most students find Reading Comprehension difficult?
ii. Why is Fact based RC the easiest one?
iii. Why is careful reading of the passage essential?

Choose the right option from the given alternatives:

iv. For answering a question on RC, one should _____
   a. be well aware of the facts, figures, etc.
   b. underline what the author is talking about in the para
   c. underline the difficult words of the passage
   d. remember the names, numbers, etc.
v. Topic based RC is tough as _____
   a. it contains technical terms
   b. it is based on universal topic
   c. it demands a lot of time and energy
   d. one can't read them fast enough
vi. A word in para '4' that means the same as - 'the process of reaching to an opinion' is _____
   a. fairly
   b. attempting
   c. inference
   d. practising.

SECTION- 'B' (Grammar and Writing Skills)

Q3 Attempt Any One of the following. [5]

a. While shopping at a popular fashionable market you were distressed to see harassment faced by some foreigners at the hands of self proclaimed guides, greedy shopkeepers, vendors and even beggars. Write a letter to the Editor, The Indian Express, mentioning that we have forgotten our culture of hospitality and are responsible for spoiling the image of a country.
b. Your locality has witnessed a number of cases of theft in recent weeks. The local law and order authorities were approached but you find no change in the situation. Write a letter in 100 to 120 words to the editor of a local newspaper drawing attention of the higher authorities to the problem. You are Ram/ Rama of 4-Govind road, Meerut.

Q4 Attempt Any Eight of the following tasks. [8]

i. Fill in the blank by using the correct form of the word in the brackets for the given portion of the letter.

Subject: Complaint against Faulty Refrigerator.

Sir

This is to respectfully submit that I write this letter with extreme disappointment that the product I _____ (receive) is faulty.

ii. Read the given sentence from a government regulation. Identify the error and supply the correction in the sentence.

'If a person does not have patience, he could not learn anything.'

Use the given format for your response.

Error	Correction

iii. Shreya sought some information from Sneha about the location of the bank. Report Shreya's question.

Shreya said, "Do you know where the nearest bank is?"

iv. Read the dialogue between Drishti and Anu.

Drishti : Where are you going?

Anu : I am going to a cafe called 'Hudson' for Arti's birthday.

Drishti asked Anu where she was going. Anu told her that _____

v. Fill in the blank to complete the sentence

He promised his father that he will become _____ (a/an/the) best student ever.

vi. Identify the error in the given sentence and supply the correction.

After using the computer for a long time, we must wash our eyes with a cold water.

Use the given format for your response

Error	Correction

vii. Select the option that identifies the error and supplies the correction for the following line.

He finally decides to call her to know how she was doing.

Option	Error	Correction
a	to	on
b	was	is
c	doing	done
d	call	calling

viii. Fill in the blank.

The Principal said that it _____ him great pleasure to meet the students that evening.

a. gives b. gave

c. was giving d. is giving

ix. Read the conversation below and complete the sentence by reporting it correctly.

"Alas! I have broken my arm," said Amit.

Amit exclaimed painfully that _____

x. I can't stand it anymore. I _____ around the furniture since breakfast.

a. am dancing b. have been dancing

c. have dancing d. was dancing

SECTION- C' (LITERATURE)

Q.5 Read the given extract and answer the questions for any one of the given two. [3]

(A) It was not only a display of pinpoint precision and military force but a demonstration of the military's loyalty to democracy, to a new government that had been freely and fairly elected. Only moments before, the highest generals of the South African defence force and police, their chests bedecked with ribbons and medals from days gone by, saluted me and pledged their loyalty.

a. What was unique about the new government?
b. Why was there a change in the attitude of the military?
c. Find a word in the extract that means 'decorated'.

(B) In my short career as a thief, I had made a study of men's faces when they had lost their goods. The greedy man showed fear, the rich man showed anger, the poor man showed acceptance. But I knew that Anil's face, when he discovered the theft, would show only a touch of sadness. Not for the loss of money, but for the loss of trust.

a. How do rich men react on being robbed?
b. Why is the speaker hesitant about robbing Anil?
c. What art had the thief developed in his short term career?

Q.6 Read the given extract and answer the questions for any one of the given two. [3]

(A) The way a crow
Shook down on me
The dust of snow
From a hemlock tree
Has given my heart
A change of mood
And saved some part
Of a day I had rued

a. What is the poet trying to appreciate through the given lines?
b. Fill in the blank with one word.
The poetic device used in the line 'and saved some part' is _____
c. What was the poet's reaction when the dust of snow fell on him?

(B) He should be lurking in shadow,
Sliding through long grass
Near the water hole
Where plump deer pass.

a. What would the tiger do in the forest?
b. Complete the following suitably. 'Shadow' here refers to the shadow of _____
c. According to the extract, why does the poet emphasise that the tiger should be 'sliding' through the foliage?

Q.7 Answer any two of the following questions in 40 to 50 words each. [6]

a. What is irony in the lesson 'A Letter to God'?
b. Write the message given by the poem 'Fire and Ice'.
c. Do you think that the seagull's family loved him? Justify their attitude towards him.

Q.8 Answer any one of the following question in 40-50 words. [3]

a. How can you say that doctor Herriot was tactful as well as full of common sense?
b. Why did Ausable not fit with any description about a secret agent?

Q.9 Answer any one of the following in 100-120 words. [5]

a. The pilot wanted to thank another pilot after his safe landing. Why? What values of the narrator are reflected from his action?
b. 'From the Diary of Anne Frank' throws light on teacher- student relationship, class atmosphere and discipline . Justify.

Q.10 Answer any one of the following in 100-120 words. [5]

a. Education is an integral factor in shaping one's life. Elucidate the statement in light of the lesson 'The Thief's Story'.
b. Mrs. Pumphrey says, "This is a Triumph of Surgery". Why did she say so? Did she know what Tricki's ailment was?

I PERIODIC TEST : 2025-26

Class - X

Subject - Artificial Intelligence (417)

Time : 2 hrs. **M.M. : 50**

General Instruction : All questions are compulsory.

Q.1- *Fill in the blanks:-* *[10 × 1 = 10]*

a. ____ is the domain of AI that deals with understanding and analyzing human language.

b. The first stage in AI Project Cycle, where aim and scope of project are defined, is called ____ .

c. In ____ type of machine learning, the data is labeled with the desired output.

d. The Artificial Neural Networks are inspired by the structure and function of ____ .

e. Delivering a speech in public is an example of ____ communication.

f. Enjoying life works as mood enhancement and releases positive hormones, like ____ .

g. The ____ is a horizontal bar usually located at the bottom of the desktop.

h. The ____ General Assembly released a global 'to-do-list', called the Sustainable Development Goals.

i. Unlabeled data is used in ____ learning.

j. It is important to make sure to ____ any programs and files that you no longer use.

Q.2- *State True or False:-* *[1 × 10 = 10]*

i. Evaluation of an AI model involves testing its performance using real-world data and relevant metrics.

ii. The Problem Scoping stage involves developing and training machine learning models.

iii. There are two sub-categories of Deep Learning; ANN and CNN.

iv. Reinforcement learning is sub-category of Machine Learning.

v. Good communication helps people understand each other better, reducing confusion and mistakes.

vi. Stress can have a negative impact on our physical health only.

vii. Email services do not have a feature to stop spam messages from reaching your email inbox.

viii. Reduced chemical use and biodiversity preservation will damage the health of the soil.

ix. Clustering is a supervised learning.

x. To rename a file, left-click on the file or folder and choose the Rename option from the context menu.

Q.3- *Multiple Choice Questions:-* *[1 × 6 = 6]*

i. Which AI application is commonly used in smart phones for voice command and personal assistance?

a) Siri
b) Google Maps
c) Camera Timer
d) Bluetooth Connectivity

ii. Clustering algorithms are commonly used in unsupervised learning for :
   a) Spam filtering b) Image classification
   c) Stock price prediction d) Grouping similar data points
iii. A message is transferred from its sender to the receiver through a ____ .
   a) Decoder b) Feedback c) Channel d) Encoder
iv. Which of the following is not an importance of working independently?
   a) Boost in confidence b) Improved Decision Making
   c) increased self-esteem d) stress Management
v. Which of the following is a single user operating system?
   a) MS DOS b) LINUX c) Mac OS d) UNIX
vi. ____ is an eco-friendly mode of transportation.
   a) Electric Vehicles b) Trucks
   c) Diesel Cars d) Airplane

Q.4- ***Write the full form of following:-*** **[1 × 4 = 4]**

i. CNN
ii. CV
iii. BIOS
iv. SDG

Q.5- ***Answer the following questions:- (Do any four)*** **[2 × 4 = 8]**

i. Describe effective communication.
ii. What is the importance of managing files and folders in computer?
iii. How can Yoga help in managing stress?
iv. Give any two examples of sustainable water practice.
v. Define DEEP Learning.
vi. Mention any two real-life applications of NLP.

Q.6- ***Answer the following questions:- (Do any four)*** **[3 × 4 = 12]**

i. What are the factors contributing to communication barriers?
ii. Write any three functions of an operating system.
iii. Sakshi is a media manager at an advertising company. She was initially very passionate about her work, but due to a lack of acknowledgment and a heavy workload, she has recently started to lose interest in it and has no motivation. What steps can she take to motivate herself to do better at work.
iv. Explain the three pillars of sustainable development.
v. Differentiate between Rule based and Learning based AI models.
vi. Match the following columns:-

Column A	Column B
a. Voice Assistant	Data Science
b. Autonomous Drone	NLP
c. Recommended System	Computer Vision

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I-PERIODIC TEST : 2025-26
CLASS - X (CBSE)
HINDI

Time: 2 hrs. M.M.: 50

निर्देश:-1. इस प्रश्न पत्र में तीन खण्ड हैं- 'क', 'ख', 'ग'।
2. सभी खण्डों के प्रश्नों के उत्तर देना अनिवार्य है।
3. दिए गए निर्देशों का पालन करते हुए सभी प्रश्नों के उत्तर दीजिए।

खण्ड 'क' (अपठित बोध)

Q.1. नीचे दिए गए गद्यांश को ध्यानपूर्वक पढ़िए और उस पर आधारित प्रश्नों के उत्तर दीजिए।

मनुष्य एक सामाजिक प्राणी है। कोई भी व्यक्ति अकेला नहीं रह सकता, क्योंकि अकेला रहना एक बहुत बड़ी साधना है। जो लोग समाज या परिवार में रहते हैं, वे इसलिए रहते हैं, क्योंकि उन्हें एक-दूसरे की आवश्यकता होती है। परिवार का अर्थ ही है माता-पिता, दादा-दादी, चाचा-चाची, भाई-बहन का समूह। इन्हीं से परिवार बनता है और कई परिवारों के मेल से समाज बनता है। व्यक्ति से परिवार, परिवार से समाज, समाज से शहर, राज्य और राष्ट्र बनते हैं। इसीलिए जो भी व्यक्ति समाज में रहता है वह एक-दूसरे से जुड़ा रहता है। मित्रता, मनुष्य के जीवन की एक अद्भुत उपलब्धि है इसीलिए मनुष्य को समाज के लिए उपयोगी बनाना पड़ता है। ईसा ने कहा है, 'जो तुममें सबसे बड़ा होगा, वह तुम्हारा सेवक होगा।' समाज की प्रवृत्ति ऐसी है कि यदि आप दूसरों के काम आएँगे, तो समय पड़ने पर दूसरे भी आपका साथ देंगे। अतः लोक सेवा से मनुष्य की यश पाने की आकांक्षा की भी पूर्ति हो जाती है। लोकसेवा के अनेक रूप हैं; जैसे- देश-सेवा, साहित्य-सेवा आदि। कोई भी रचनात्मक कार्य, जिससे सार्वजनिक हित हो, वह परोपकार है। रोग, मृत्यु, शोक, बाढ़, भूकंप, संकट, दरिद्रता, महामारी, उपद्रव आदि कदम-कदम पर सहानुभूति दर्शाने के अवसर विद्यमान हैं। सहानुभूति और परोपकार का अर्थ केवल दधीचि और गाँधी जैसा बलिदान ही नहीं, बल्कि दयालुता के छोटे-छोटे कार्य, मृदुता का व्यवहार, पड़ोसियों का सहायक बनने की चेष्टा, दूसरों की भावनाओं को ठेस न पहुँचाना, दूसरों की दुर्बलताओं के प्रति उदार होना, किसी को निर्धन न मानना आदि भी सहानुभूति से जुड़े परोपकार के ही लक्षण हैं।

(i) गद्यांश के आधार पर लोकसेवा का कौन-सा रूप है ? [1]
(क) देश की सेवा (ख) साहित्य की सेवा
(ग) 'क' और 'ख' दोनों (घ) सार्वजनिक हित

(ii) समाज का निर्माण किसके मेल से होता है ? [1]
(क) माता-पिता के मेल से (ख) लड़ाई-झगड़े से
(ग) कई परिवारों के मेल से (ग) ये सभी

(iii) निम्नलिखित कथन को पढ़कर सही विकल्प का चयन कीजिए - [1]
समाज में कोई व्यक्ति अकेला नहीं रह सकता, क्योंकि
(i) मनुष्य एक सामाजिक प्राणी है।
(ii) अकेला रहना बहुत बड़ी साधना है।
(iii) अकेला मनुष्य भयभीत रहता है।
(iv) अकेला रहना असंभव कार्य है।
कूट :-
(क) केवल (i) सही है
(ख) (i) और (ii) सही है
(ग) (ii) और (iii) सही है
(घ) (iii) और (iv) सही है

(iv) 'लोक सेवा से यश पाने की आकांक्षा की पूर्ति भी हो जाती है।' पंक्ति का आशय स्पष्ट कीजिए। [2]

Q2. नीचे दिए गए काव्यांश को ध्यान पूर्वक पढ़िए और उस पर आधारित प्रश्नों के उत्तर दीजिए -

नीलांबर परिधान हरित पट पर सुंदर है,
सूर्य चंद्र युग-मुकुट, मेखला रत्नाकर है,
नदियाँ प्रेम-प्रवाह, फूल तारे मंडल हैं,
बंदीजन खग-वृंद, शेषफल सिंहासन है
[illegible]
जिसके कारण धूल भरे हीरे कहलाए,

हम खेले कूदे हर्षयुत, जिसकी प्यारी गोद में
हे मातृभूमि! तुझको निरख, मग्न क्यों न हो गोद में
निर्मल तेरा नीर अमृत के सम उत्तम है,
शीतल मंद सुगंध पवन हर लेता श्रम है,
षट्ऋतुओं का विविध दृश्य युत अद्भुत क्रम है,
हरियाली का फर्श नहीं मखमल से कम है
करते अभिषेक पयोद हैं, बलिहारी इस वेश की
हे मातृभूमि! तू सत्य ही, सगुण मूर्ति सर्वेश की,
जिसकी रज में लोट-लोटकर बड़े हुए हैं,
घुटनों के बल सरक-सरककर खड़े हुए हैं,
शुचि-सुधा सींचता रात में तुझ पर चंद्रप्रकाश है
हे मातृभूमि! दिन में तरणि, करता तम का नाश है
जिस पृथ्वी में मिले हमारे पूर्वज प्यारे,
उससे है भगवान! कभी हम रहें न न्यारे,
लोट-लोटकर वहीं हृदय को शांत करेंगे,
उसमें मिलते समय मृत्यु से नहीं डरेंगे,
उस मातृभूमि की धूल में, जब पूरे सन जाएँगे
होकर भव-बंधन-मुक्त हम, आत्मरूप बन जाएँगे।

(i) कवि अपना मन किस प्रकार शांत करना चाहता है ? [1]

(क) जिस भूमि पर कवि का बचपन बीता था वहाँ रहकर।
(ख) जिस भूमि पर कभी उसके पूर्वज रहते थे, उस मिट्टी में लोटकर।
(ग) हरे-भरे स्थानों पर कुछ निवास कर।
(घ) किसी एकांत स्थान पर निवास कर।

(ii) नीलांबर परिधान हरित पट पर सुंदर है' में हरित पट का अर्थ है – [1]

(क) चारों ओर फैले हरे-भरे पेड़ पौधे।
(ख) हवा में लहराते हरे रंग के वस्त्र।
(ग) चारों ओर फैली हरियाली से युक्त धरती।
(घ) नीला वस्त्र पहने को सुंदर स्त्री।

(iii) निम्नलिखित कथन पढ़कर सही विकल्प का चयन कीजिए। [1]

कवि अपनी मातृभूमि पर बलिहारी होना चाहता है, क्योंकि –

(i) अपनी मातृभूमि के प्रति अतिशय प्रेम रखता है।
(ii) अपनी मातृभूमि पर फैली हरियाली से प्रेम करता है।
(iii) वह मातृभूमि पर बहने वाली सुगन्धित हवा से प्रसन्न रहता है।
(iv) अपनी मातृभूमि को प्रतिक्षण याद करता है।

कूट :-

(क) केवल (i) सही हैं।
(ख) (ii) और (iii) सही हैं।
(ग) (i), और (iii) सही हैं।
(घ) (i), (ii), (iii), व (iv) सही है।

(iv) प्रस्तुत काव्यांश का केंद्रीय भाव स्पष्ट कीजिए। [2]

खण्ड–'ख' (व्यावहारिक व्याकरण)

Q.3. निर्देशानुसार 'वाक्य भेद' पर आधारित पाँच प्रश्नों में से किन्हीं चार प्रश्नों के उत्तर दीजिए– [1×4=4]

(क) तुम घर गए और वह रोने लगी। सरल वाक्य में बदलिए।
(ख) घंटी बजी, छात्र पुस्तकें लेकर कक्षा से बाहर निकले। संयुक्त वाक्य में बदलिए।
(ग) राम के आने पर सभी घर की ओर चल दिए। मिश्र वाक्य में बदलिए।
(घ) लाभ वाला काम करो। मिश्र वाक्य में बदलिए।
(ङ) सभ्यता साधन है जबकि संस्कृति साध्य है। वाक्य का भेद बताइए।

Q4. निर्देशानुसार 'वाच्य' पर आधारित पाँच प्रश्नों में से किन्हीं चार प्रश्नों के उत्तर दीजिए :– [1×4=4]

(क) बच्चों से चुप नहीं बैठा जाता। वाच्य का भेद बताइए।
(ख) अध्यापक ने हमें नया पाठ पढ़ाया। कर्मवाच्य में बदलिए।
(ग) पुलिस द्वारा लाठीचार्ज किया गया। कर्तृवाच्य में बदलिए।
(घ) मोहन चला नहीं सकता। भाववाच्य में बदलिए।
(ङ) माँ ने भिखारी को भोजन दिया। कर्मवाच्य में बदलिए।

Q5. निर्देशानुसार अलंकार पर आधारित पाँच प्रश्नों में से किन्हीं चार प्रश्नों के उत्तर दीजिए :– [1×4=4]

(क) पानी परात को हाथ छुयो नहि नैनन के जल सो पग धोए। पंक्ति में कौन सा अलंकार है।
(ख) 'लघु तरनि हरिनी–सी सुन्दर' पंक्ति में कौन सा अलंकार है।
(ग) उत्प्रेक्षा अलंकार का उदाहरण लिखिए।
(घ) 'गोपी पद–पंकज पावन कि रज सिर भीजै।' पंक्ति में अलंकार है।
(ङ) मेघमय आसमान से उतर रही है संध्या सुंदरी परी सी धीरे–धीरे।
काव्य पंक्ति में कौन सा अलंकार है।

Q6. निम्नलिखित पठित गद्यांश पर आधारित बहुविकल्पीय प्रश्नों के सर्वाधिक उपयुक्त विकल्प चुनकर लिखिए। [1×5=5]

पूरी बात तो अब पता नहीं, लेकिन लगता है कि देश के अच्छे मूर्तिकारों की जानकारी नहीं होने और अच्छी मूर्ति की लागत अनुमान और उपलब्ध बजट से कहीं बहुत ज्यादा होने के कारण काफी समय ऊहापोह और चिट्ठी–पत्री में बरबाद हुआ होगा और बोर्ड की शासनावधि समाप्त होने की घड़ियों में किसी स्थानीय कलाकार को ही अवसर देने का निर्णय किया होगा और अंत में करबे के इकलौते हाई स्कूल के इकलौते ड्राइंग मास्टर मान लीजिए, मोतीलाल जी को ही यह सौंप दिया गया होगा, जो महीने भर में मूर्ति बनाकर पटका देने का विश्वास दिला रहे थे।

(क) गद्यांश के आधार पर बताइए कि मूर्ति बनाने का अवसर किसे दिया गया ?

(i) स्थानीय कलाकार को (ii) लेखक के सहयोगी को
(iii) सरकारी कार्यालयों को (iv) विदेशी कलाकारों को

(ख) मूर्ति बनाने के काम में क्या-क्या परेशानियाँ आई होंगी ?

(i) अच्छे मूर्तिकारों का अभाव
(ii) उपलब्ध बजट का कम होना
(iii) (i) और (ii) दोनों
(iv) स्थानीय कलाकार का गुणी होना

(ग) मोतीलाल जी ने मूर्ति बनाने का कार्य कब पूरा करने का विश्वास दिलाया ?

(i) महीने भर में (ii) हफ्ते भर में
(iii) सप्ताह भर में (iv) इनमें से कोई नहीं

(घ) स्थानीय कलाकार को मूर्ति बनाने का कार्य क्यों सौंपा गया ?

(i) अच्छे मूर्तिकारों की जानकारी के अभाव में
(ii) नगरपालिका बोर्ड की शासनावधि समाप्त होने के कारण
(iii) (i) और (ii) दोनों
(iv) अच्छी मूर्ति बनाने के लिए

(ङ) मूर्ति बनाने में काफी समय ऊहापोह तथा चिट्ठी-पत्री में क्यों बरबाद हुआ ? उचित विकल्प का चयन कीजिए।

(i) अच्छी मूर्ति की लागत अनुमान व उपलब्ध बजट से ज्यादा होने के कारण
(ii) स्थानीय कलाकारों की ज्यादा उपलब्धता होने के कारण।
(iii) मुख्य मूर्तिकारों की ज्यादा उपलब्धता होने के कारण।
(iv) विदेशों से बुलाए गए मूर्तिकारों के कारण।

कूट –

(क) केवल कथन (i) सही है।
(ख) केवल कथन (ii) सही है।
(ग) कथन (iii) और (iv) सही है।
(घ) कथन (i) और (ii) सही है।

[2×3=6]

Q.7. निम्नलिखित में से किन्हीं तीन प्रश्नों के उत्तर लिखिए –

1. कैप्टन कौन था ? वह मूर्ति के चश्मे को बार-बार क्यों बदल दिया करता था ?
2. पानवाले का रेखाचित्र प्रस्तुत कीजिए।
3. लखनवी अंदाज पाठ के लेखक ने सेकंड क्लास का टिकट क्यों लिया ?
4. नवाब साहब ने खीरा खाने की पूरी तैयारी की, बाद में उसे बिना खाए फेंक दिया। इस नवाबी व्यवहार पर टिप्पणी कीजिए।

खण्ड– ग

[1×5=5]

Q.8. निम्नलिखित काव्यांश को ध्यानपूर्वक पढ़कर पूछे गए प्रश्नों के उचित उत्तर दीजिए –

उसकी स्मृति पाथेय बनी है थके पथिक की पंखा की।
सीवन को उधेड़कर देखोगे क्यों मेरी कंथा की ?
छोटे से जीवन की कैसे बड़ी कथाएँ आज कहूँ ?
क्या यह अच्छा नहीं कि औरों की सुनता मैं मौन रहूँ ?
सुनकर क्या तुम भला करोगे मेरी भोली आत्मकथा ?
अभी समय भी नहीं, थकी सोई है मेरी मौन व्यथा।

1. 'स्मृति को पाथेय' बनाने का अभिप्राय है –
   (क) निराशा में आशा ढूँढना
   (ख) हताशा और निराशा
   (ग) अतीत की सुखद स्मृतियों के सहारे शेष जीवन व्यतीत करना
   (घ) इनमें से कोई नहीं
2. काव्यांश में थका पथिक कौन है ?
   (क) स्वयं कवि
   (ख) मार्ग पर जाता हुआ पथिक
   (ग) अतीत की स्मृतियाँ
   (घ) उपर्युक्त सभी
3. यहाँ 'कंथा' शब्द का प्रयोग किसके लिए हुआ है ?
   (क) फटा हुई दरी के लिए
   (ख) कंधे के लिए
   (ग) फटी गुदड़ी के समान जर्जरित जीवन के लिए
   (घ) निराश के मन के लिए
4. 'अभी समय नहीं है' कवि के अनुसार किस बात का समय नहीं है ?
   (क) कविता लिखने का
   (ख) कवि के हृदय की सुप्त व्यथाओं को प्रकट करने का
   (ग) लोगों द्वारा दिए गए धोखे के बारे में बताने का
   (घ) उपर्युक्त सभी
5. काव्यांश में 'सीवन को उधेड़कर' देखने का क्या आशय है ?
   (क) विस्मृति पीड़ा को कुरेदना
   (ख) अपने बारे में सबको बताना
   (ग) अपने सुखद दिनों को याद करना
   (घ) इनमें से कोई नहीं

[2×3=6]

Q.9. निम्नलिखित में से किन्हीं तीन प्रश्नों के उत्तर दीजिए –

(क) कवि आत्मकथा लिखने से क्यों बचना चाहता है ?
(ख) उद्धव के व्यवहार की तुलना किस-किस से की गई है ?
(ग) नरजादा न रहीं के माध्यम से कौन सी मर्यादा न रहने की बात की जा रही है ?
(घ) संगतकार किन-किन रूपों में मुख्य गायक की मदद करते हैं ?

[3×2=6]

Q.10. निम्नलिखित में से किन्हीं दो प्रश्नों के उत्तर दीजिए :–

(क) पिता द्वारा भोलानाथ को खाना खिलाने के बाद भी उसकी माँ उसे खाना खिलाती थी क्यों ?
(ख) माता का आँचल पाठ के आधार पर कहा जा सकता है कि बच्चे का अपने पिता से अधिक जुड़ाव था, फिर भी विपदा के समय वह पिता के पास न जाकर माँ की शरण लेता है। आपकी समझ से इसकी क्या वजह हो सकती है ?
(ग) भोलानाथ बचपन में कैसे-कैसे नाटक खेल करता था ? इससे उसका कैसा व्यक्तित्व उभरकर आता है ?

I PERIODIC TEST : 2025-26
CLASS – X (CBSE)
SCIENCE [086]

Maximum Marks : 50
Time allowed : Two hours

General Instructions : -

- This question paper comprises of four Sections - A, B, C and D.
- All questions are compulsory.
- Section - A consists of 1 mark questions, it contains MCQ & case study type questions.
- Section - B consists of 1 mark questions, it contains very short answer and Assertion - Reason type questions.
- Section - C consists of 2 marks questions, it contains short answer type questions.
- Section - D consists of 5 marks questions, it contains long answer type questions.

SECTION-A [20 Marks]
COMPETENCY BASED QUESTIONS (MCQ and Case Based Type Questions)

Question 1

The vice box in humans is called: [1]

(a) Pharynx (b) Larynx
(c) Trachea (d) Alveoli

Question 2

In the nephron of man, what is the correct sequence of urine formation? [1]

(a) Selective reabsorption, tubular secretion and ultrafiltration
(b) Ultrafiltration, tubular secretion and selective reabsorption
(c) Ultrafiltration, selective reabsorption and tubular secretion
(d) Tubular secretion, selective reabsorption and ultrafiltration

Question 3

The characteristic processes observed in anaerobic respiration are [1]

(A) lack of oxygen
(B) release of carbon dioxide
(C) Release of low Amount of energy
(D) Release of lactic acid and ethyl alcohol

(a) A & D (b) B, C & D
(c) A, B, & D (d) A, B, C & D

Question 4

Which of the following is/are olfactory indicator/s. [1]

(a) Vanilla (b) Clove
(c) Litmus (d) Onion
(i) Only (a) & (b) (ii) All (a), (b) (c) & (d)
(iii) Only (c) (iv) (a), (b) & (d)

Question 5

Four solutions namely glucose, alcohol, hydrochloric acid and sulphuric acid filled in four separate beakers are connected one by one in an electric circuit with a bulb. The solutions in which the bulb will glow when current is passed are : [1]

(a) Glucose and alcohol
(b) Alcohol and hydrochloric acid
(c) Glucose and sulphuric acid
(d) Hydrochloric acid and sulphuric acid

Question 6

Reena took 5 ml of Lead Nitrate solution in beaker and added approximately 4 ml of Potassium Iodide solution to it. What would she observe? [1]

(a) The solution turned red
(b) Yellow precipitate was formed
(c) White precipitate was formed
(d) The reaction mixture became hot

Question 7

Under which of the following condition can a concave mirror form an image which is real, inverted and larger than the actual size of the object? [1]

(a) When the object is kept at a distance less than its focal length.
(b) When the object is kept between the center of curvature and the focus.
(c) When the object is kept at the center of curvature.
(d) When the object is kept beyond the center of curvature.

Question 8

Which property of light remains unchanged during refraction? [1]

(a) Speed (b) Frequency
(c) Wavelength (d) Direction

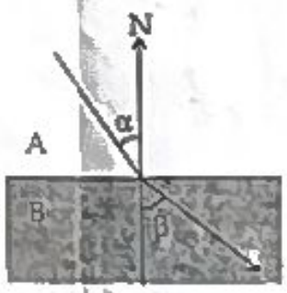
Question 9

Which of the following lens would you prefer to use while reading small letters? [1]

(a) A convex lens of focal length 50 cm.
(b) A concave lens of focal length 50 cm.
(c) A convex lens of focal length 5 cm.
(d) A concave lens of focal length 5 cm.

Question 10

A light ray enters from medium A to medium A to medium B as shown in the figure. [1]



The relative refractive index of medium A relative to medium B will be-

(a) greater than unity (b) less than unity
(c) equal to unity (d) zero

[CASE STUDY-BASED QUESTIONS (Q. 11 TO Q.13)] [1x4=4]

Question 11

Read the following passage and answer the questions that follows:

Sumit's father was suffering from severe stomach pain and irritation. He was showing symptoms of heart burn and indigestion. He consulted a doctor who told him that he was suffering from acidity and advised him to take an antacid tablet after each meal. Sumit's father followed the advice strictly and was cured.

Q.11A. Which acid is produced in our stomach? [1]

(a) Sulphuric acid (b) Hydrochloric acid
(c) Nitric acid (d) Phosphoric acid

Q.11B. Antacids contain: [1]

(a) acid (b) base
(c) salt (d) both (a) and (b)

Q.11C. Milk of magnesia is an antacid which contains: [1]

(a) magnesium chloride (b) magnesium phosphate
(c) magnesium hydroxide (d) magnesium nitrate

Q.11D. Identify the type of reaction involved in it [1]

(a) combination reaction (b) neutralisation reaction
(c) decomposition reaction (d) single displacement reaction

Question 12

Read the following and answer the questions that follow. [1x3=3]

Sudha finds out that the sharp image of the window of her science laboratory is formed at a distance of 15 cm from the lens. She now tries to focus the building visible outside the window on her screen without disturbing the lens thus moving the screen to form a sharp image on the screen.

(a) Which lens is Sudha using? [1]
(b) What is the nature of the image formed? [1]
(c) Which direction should she move the screen to form a sharp image while trying to focus the building? [1]

Question 13

Read the following paragraph and answer the following questions. [3]

A group of students is conducting an experiment to study the rate of photosynthesis in a potted plant. They placed identical plants under three different condition:

- Plant A : Well-lit area with Sufficient water and Carbondioxide.
- Plant B : Dimly lit area with Sufficient water and Carbondioxide.
- Plant C : Well-lit area with limited water and Sufficient Carbondioxide.
- After a few hours, they test the leaves of each plant for the presence of starch.

(a) In which plant (A, B or C) would the presence of starch be most significant?
(b) Write the balanced Equation for the process studied.
(c) Explain the role of stomata in the leaves during photosynthesis.

SECTION-B [10 Marks]
Objective Type Question

Question 14

While diluting an acid, why is it recommended that the acid should be added to water and not water to the acid? [1]

Question 15

Give one example of thermal decomposition in the form of balanced equation. [1]

Question 16

(i) The linear magnification produced by a mirror is "+1". State the nature of image so formed & identify the mirror used. [1]

(ii) Refractive index of flint glass is 1.65 and that for alcohol is 1.36 with respect to air. What will be the refractive index of flint glass with respect to alcohol? [1]

Question 17

Write one specific function of each of the following in relation to human excretory system. [1×3=3]

(a) Urethra
(b) Glomerulus
(c) Nephron

[ASSERTION AND REASON TYPE QUESTIONS]

Question no. 18 to 20 consists of two statements-Assertion (A) and Reason(R)- Answer these questions selecting the appropriate option given below :

Option (a) if both Assertion and Reason are true and Reason is the correct explanation of Assertion.

Option (b) if both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

Option (c) if only Assertion is true but Reason is false.

Option (d) if Assertion is false but reason is true. [1]

Question 18

Assertion(A) : Ferrous sulphate crystals are green in colour but on heating, they first turn white and on further heating decompose to leave behind a reddish-brown residue.

Reason (R) : They turn white due to loss of water of crystallization and the reddish - brown residue is due to formation of Fe_2O_3. [1]

Question 19.

Assertion (A): The reflector of search-light are made up of concave spherical mirror.

Reason (R) : The concave spherical mirror diverges the light of the bulb which is placed at its focus thus covering a greater area. [1]

Question 20

Assertion(A) : Bile juice is important for digestion of fats.

Reason (R) : Bile juice is acidic in nature.

SECTION - C [10 Marks]
[SHORT ANSWER TYPE QUESTIONS]

Question 21

A convex lens of focal length 10 cm can produce a virtual erect as well as a real inverted image. [2]

(a) Is this statement correct?

(b) What will be the nature and size of the image when the object is placed 5 cm in front of lens?

Question 22

(A) Solutions X, Y and Z have pH values 1, 9 and 10 respectively. Arrange them in increasing order of basic character giving reasons. [1+1=2]

OR

(B) What does one mean by exothermic and endothermic reactions? Give examples. [1+1=2]

Question 23

(A) Ananya heated a dish containing copper powder & got different observations. Write one prominent observation what she must have observed. Also Write the reason for that observation. [1+1=2]

OR

(B) Write the balanced equation for the following chemical reactions. [1+1=2]

(i) Hydrogen+Chlorine → Hydrogen chloride.

(ii) Barium chloride+Aluminum sulphate → Barium sulphate+Aluminum chloride $H_2 + Cl_2 \longrightarrow 2HCl$

Question 24

How do plants obtain food? What are the two phases of Photosynthesis? [2]

Question 25

2 g of silver chloride is taken in a China dish and the China dish is placed in sunlight for some time. What will be the observation in this case? Write the chemical reaction involved in the form of a balanced chemical equation. Identify the type of chemical reaction. [½+1+½=2]

SECTION - D [2×10 Marks]
[LONG ANSWER TYPE QUESTIONS]

Question 26

Size of the virtual image formed by a mirror having focal length 20 cm is observed to be reduced to 1/3rd of object size.

(a) At what distance has the object been placed? (Show calculation) [2]

(b) Which mirror is it? [1]

(c) Draw a ray diagram for the above case. [1]

(d) Write any two uses of the mirror. [1]

OR

Size of the virtual image formed by a lens of the object placed 20 cm in front of it, is observed to be reduced to half of the object size.

(a) Which lens is it? [1]

(b) Find the focal length and the power of the lens. (Show calculation) [2]

(c) Draw a ray diagram for the above case. [2]

Question 27

(A) Draw the diagram of alimentary canal of man and label the following parts. [5]

(i) Pharynx (ii) Oesophagus

(iii) Duodenum (iv) Large intestine.

(B) Answer the following questions-

(a) How do leaves of plants help in excretion? Through

(b) Name the correct substrates for the following enzymes.

(i) Trypsin - Carb (ii) Amylase - fat

(iii) Pepsin - P (iv) Lipase - Fat

(c) What is common for Cuscuta, ticks and leeches? Parasitic Nutrition

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